From Pixels to Concepts: Enhancing Interpretability of Image Classifiers with Multimodal Language Models

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Existing explainability methods for image classifiers, such as heatmaps or feature attributions, often lack semantic clarity and provide limited support for concept-level inspection by users. We introduce TEBOL (Text-Based Embedding for Object Localization), a method for interpretable image classification that explains predictions through semantic concepts rather than visual saliency. TEBOL prompts a multimodal Large Language Model (LLM) to describe an image in natural language, embeds each word, and averages the embeddings into a sentence-averaged embedding (SAVe) on which a logistic regression classifier is trained. Because the classifier is linear over a mean, SMER decomposes its output exactly: the per-word contributions reconstruct the model's own logit to within 5×10^{-7}, at the cost of one dot product per word and with no sampling. The highest-ranked words are then grounded by prompting a multimodal LLM for bounding boxes, which should be interpreted as post-hoc semantic grounding rather than a direct visualization of the classifier's internal spatial attention. We evaluate TEBOL on 14,351 ImageNet images spanning ten object classes, in a closed-weight configuration (GPT-4o) and an open-weight one (Qwen3-VL-32B-Instruct), both under 5-fold cross-validation repeated five times, the GPT-4o configuration on four of the five tasks. Image-level accuracy is 0.941 to 0.998 across five binary tasks, against 0.952 for a CLIP linear probe on the same images. Removing the class name and its synonyms from the descriptions costs 0.123 to 0.239 of accuracy while leaving it far above chance, so the classifier does not merely read the label out of the text. The ranked words localize the annotated object with a mean intersection over union of 0.829, and a perturbation analysis finds SMER and LIME equally faithful, their areas over the perturbation curve differing by at most 0.001 on all five tasks. A preregistered user study with 270 participants found TEBOL's concept-grounded bounding boxes more comprehensible overall than Grad-CAM heatmaps ($p < 0.001$), while ratings of how well each display captures the named object did not differ between formats. Taken together, these results suggest that language-aligned explanations can support perceptually grounded and human-readable explainable computer vision.

CCS Concepts: • **Human-centered computing** → **Visualization design and evaluation methods**; • **Computing methodologies** → *Machine learning*; Object detection.

Additional Key Words and Phrases: human perception, explainable AI, image classification, multimodal language models, concept-based explanations, saliency maps, human-subject evaluation

1 Introduction

As machine learning models grow increasingly complex, the main bottleneck in human-AI collaboration has shifted from algorithmic accuracy to human perceptual comprehension. In computer vision, explanations of classifier decisions are still dominated by pixel-level saliency maps such as Grad-CAM [19] and LIME [18]. These methods highlight predictive regions through continuous color gradients, but from a perceptual standpoint they often remain fragmented, overlapping, and semantically ambiguous. As a result, users may struggle to map such heatmaps onto the discrete object- and concept-level structures through which humans typically parse scenes [1, 13].

Recent discourse in applied perception underscores the importance of aligning computational visual models with human visual attention and perceptual constraints. For example, Zhang et al. showed that saliency-guided generation benefits from incorporating human gaze priors directly into the image formation pipeline [26]. Likewise, Surace et

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al. demonstrated that foveated reconstruction better preserves perceptually important image features than purely statistical criteria [22]. While these studies apply perceptual priors to generation and rendering, they point to a broader principle that is also relevant for explainable AI: explanations should be designed for human perceptual processing rather than only for algorithmic convenience.

To address this perceptual disconnect, we introduce TEBOL (Text-Based Embedding for Object Localization), an explainability pipeline that translates opaque visual evidence into cognitively manageable semantic concepts. Instead of relying on continuous spatial heatmaps, TEBOL uses a multimodal Large Language Model (LLM) to produce textual descriptions, a transparent additive classifier to derive exact word-level attributions, and a multimodal grounding step to visualize the most important concepts using discrete bounding boxes. This design presents explanations in an object-oriented form that better matches how people parse scenes. Caption length is a design parameter rather than a tuned one: we set it initially at seven words, following classic and contemporary work on bounded working memory and cognitive load [2, 12], and then measure accuracy across eight lengths from three to thirty words.

We evaluate TEBOL along three axes: classification accuracy, the faithfulness of its word-level attribution, and how human observers perceive and prefer the resulting visualizations. Because the pipeline's accuracy depends on what the caption happens to contain, we additionally measure how often a caption names the class it describes, and repeat the classification and attribution experiments with those terms and their synonyms removed. This design follows recent calls for explainable AI to support epistemic transparency and scientific understanding [9, 10].

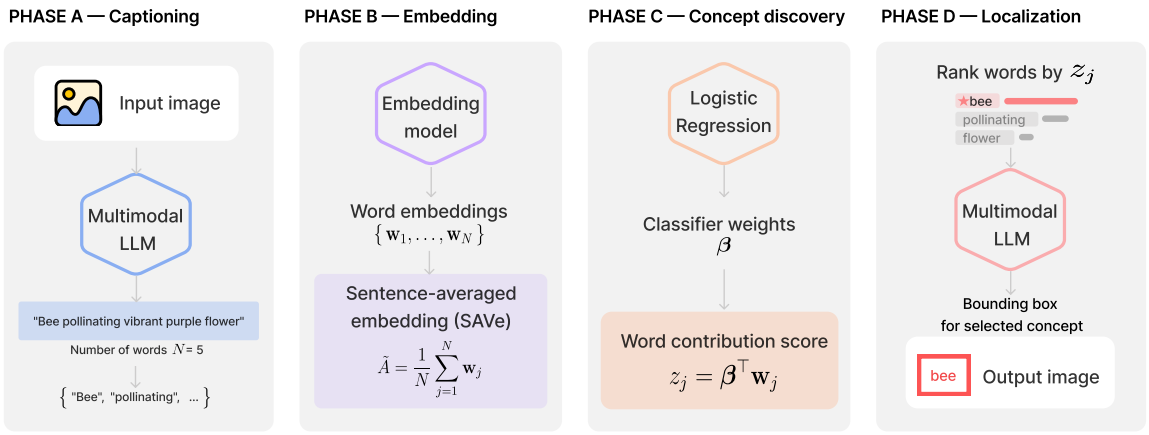


Fig. 1. Workflow of the proposed method for simultaneous image classification and explanation using LLMs, shown on the worked example of Section 3.1. In Phase A a multimodal LLM describes the image in n words and the description is split into its words. In Phase B each word is embedded and the embeddings are averaged into the sentence-averaged embedding. In Phase C logistic regression is fitted on that vector and each word receives a contribution score $z_j = \beta^\top \mathbf{w}_j$. In Phase D the highest-ranked words are passed to a multimodal LLM, which returns a bounding box for the selected concept.

1.1 Overview of the proposed method.

Our approach, illustrated in Figure 1, starts by generating concise textual descriptions of images using a multimodal LLM. Each word of the caption is embedded and the embeddings are averaged into a single vector, the SAVe (Sentence Averaged Embedding). Averaging is the operation that makes the later attribution exact; it also discards word order and splits multi-word concepts, and we return to both costs in Section 3.2.

We then train a logistic regression model to map the SAVe-based sentence embeddings to image class labels. This combination of sentence-averaged embeddings and an additive classification model enables the use of the SMER (Self Model Entities Rated) technique [6], which decomposes the model's output into additive importance scores for each individual word.

Once the most important words are identified, we use them to locate and draw bounding boxes around relevant regions in the images. This is done by prompting a multimodal LLM [15] with the selected keywords to return bounding box coordinates. The class name is never included in this prompt: only the words SMER ranked highest are passed, so a box is requested for the concept the classifier used rather than for the label. We evaluate the resulting boxes against ImageNet's own annotations in Section 6.6.

The strongest faithfulness claim in TEBOL applies to the textual attribution stage. SMER is faithful to the logistic regression classifier, whereas the bounding boxes are produced by an external multimodal LLM and should therefore be interpreted as post-hoc semantic grounding of the selected concepts rather than a direct map of where the classifier "looked" in the image.

The choice of logistic regression over averaged embeddings is deliberate. Logistic regression offers full transparency and allows explanation through SMER—which would not be feasible with more complex black-box models. Moreover, recent research has shown that logistic regression can achieve a favorable trade-off between interpretability and accuracy, especially in high-stakes domains like healthcare [17, 20].

1.2 Progress against the state-of-the-art.

As outlined in Figure 1, our TEBOL method performs explainable image classification by combining a standard logistic regression classifier with an LLM-based explanation module. We use a simple and transparent model, while explanations are generated through SMER—a newly proposed algorithm whose word-level contributions reconstruct the model's own logit to within 5×10^{-7}. This eliminates the need for post-hoc explanation methods at the textual attribution stage, which are often approximate and inconsistent. We then complement this faithful concept attribution with bounding boxes generated by an external multimodal LLM, yielding a form of semantic grounding that is especially suitable for human inspection. The individual components are established. What is new is their combination: exact word-level attribution over generated captions, grounded back into the image, and evaluated under explicit label-leakage controls.

We conducted a comparative evaluation against two widely used explanation methods in computer vision: LIME (Local Interpretable Model-agnostic Explanations) [18] and Grad-CAM (Gradient-weighted Class Activation Mapping) [19]. This evaluation examines the interpretability and communicative usefulness of language-based, concept-grounded explanations relative to traditional pixel-based techniques, while acknowledging that these methods do not visualize the same underlying explanatory object. For the classification step, we compare against contrastively pretrained vision–language models—CLIP [16] and CoCa [25]—both as zero-shot classifiers and as linear probes on frozen image features, together with direct classification by the captioning model itself.

We ran a preregistered user study (N = 270) comparing the interpretability of the bounding boxes generated by our method with heatmaps, which are commonly used with neural image classifiers. Such a comparison enables empirical evaluation of user understanding and trust, and tests whether concept-grounded outputs are better aligned with human perceptual processing.

1.3 Research objectives and contributions.

Building on this motivation, we aimed to design an end-to-end image classification pipeline, which is:

(1) grounded in natural language concepts, where 'interpretable' means providing explanations that are faithful to the classifier's decision-making process at the word level, aligned with human-understandable semantic concepts (keywords), and augmented with visual concept grounding (via bounding boxes);
(2) evaluated both quantitatively, by comparing semantic word-level attributions and visual bounding boxes with established methods, and qualitatively, through a user study;
(3) separable from the information the caption already carries, so that the pipeline's contribution can be measured independently of whether the caption names its own class.

The contributions of this work are as follows:

(1) We introduce TEBOL, an inherently interpretable NLP-based pipeline for image classification ensuring exact word-level attribution of predictions to semantic concepts.
(2) We ground the highest-ranked concepts back into the image and evaluate the resulting boxes against ImageNet annotations, reaching a mean IoU of 0.76 to 0.91.
(3) We measure label leakage and its cost: 87% of generated captions name the class they describe, and removing those terms costs up to 0.24 accuracy.
(4) We report a preregistered study with $N = 270$ participants in which bounding boxes were rated more interpretable than Grad-CAM heatmaps overall, with per-category results mixed.

2 Related work

Recent work on LLM-based image classification integrates visual inputs with textual representations to leverage cross-modal semantic information.

2.1 Comparison with related LLM-based image classifiers

The most closely related approach uses an LLM to generate descriptive textual features, which then serve as input to a downstream classifier [11]. There, descriptors are generated from the class name by a text-only model, so they are identical for every image of a class and the explanation reports which generic class attributes matched. In TEBOL the words are generated from the image itself, so the attribution is specific to the photograph, and the class name is never supplied to the captioner. TEBOL also grounds the selected concepts spatially, which descriptor-matching does not provide.

LLaMP [27] tackles low-shot image classification by reframing the task as a language modeling problem. In this method, the image content and class labels are embedded into prompts, and the LLM predicts the correct label as the next token. This formulation allows classification with limited training data and without task-specific fine-tuning [27]. However, because the LLM itself acts as the classifier, no intermediate representation is available for introspection or interpretation. Unlike these methods, our pipeline separates classification and attribution, enabling concept-level interpretation of predictions derived from LLM-generated captions.

2.2 Comparison with feature-based methods for explaining image classifiers

Several approaches have proposed feature-based explanation methods to interpret image classifier predictions. An example is LIME, which perturbs the input and identifies the regions that most affect the model's predictions [18].

Attribution methods such as Grad-CAM have been reported to exhibit variability under different evaluation settings [5, 14], and several post-hoc techniques provide limited fidelity with respect to the underlying model behavior [14]. TEBOL derives its attribution from the classifier's parameters directly, so no sampling is involved and the decomposition is exact at any caption length.

2.3 Perceptual processing of CAM-based vs. concept-based visual explanations

Grad-CAM [19] and related approaches such as ScoreCAM [21] visualize salient image regions through gradient-derived heatmaps. From the standpoint of human perception, however, these visualizations often remain diffuse, continuous, and only loosely aligned with object boundaries. Prior work has shown that saliency maps may remain visually similar under certain perturbations, raising concerns about their sensitivity to model-relevant features [1]. A heatmap also provides no semantic anchoring: a continuous gradient indicates where the model responded, but does not name what is there.

Human visual processing depends on spatial structure and discrete feature boundaries, and visual systems built for people perform better when they respect those constraints rather than relying on low-level numerical importance signals alone [22, 26].

Applying these perceptual insights to explainable AI implies that useful explanations should be structured around semantically meaningful, visually bounded units. This is consistent with work arguing that concept-based explanations provide a more interpretable alternative to unstable feature-attribution methods [3], and with evidence that explanations do not reliably improve understanding or trust when they lack alignment with natural human reasoning [23]. Our approach follows this direction by replacing heatmaps with concept-grounded bounding boxes, each tied to a named concept. We test whether that form is easier for people to interpret in Section 7.3.

2.4 Ensuring fidelity in model explanations through concept-based method

Our pipeline employs SMER [6], a method for quantifying word-level importance in linear models. Unlike LIME [18], which approximates model behavior using stochastic sampling and surrogate models, SMER enables analytical decomposition of a logistic regression model into additive word-level contributions. Because the rearrangement is algebraic, the per-word contributions sum back to the model's own output exactly.

We add a downstream step: the highest-ranked words are passed to a multimodal model that returns bounding boxes for them in the original image.

3 Method

This section describes the workflow of the proposed method, following the sequence illustrated in Figure 1.

3.1 Representing image content with text using LLMs

Our approach begins by generating concise textual descriptions for images. The prompt is a single user message with no system prompt: *"Describe this image in {n} words."* We report two configurations of the pipeline (Table 1). The GPT-4o configuration uses GPT-4o as the captioner at seven words, one caption per image. The Qwen3-VL configuration uses the open-weight Qwen3-VL-32B-Instruct and varies n over eight values—3, 5, 7, 10, 15, 20, 25 and 30—with five repetitions at each, at temperature 1.0. The method does not depend on which model produces the caption; we state which configuration each result comes from.

The initial choice of seven words followed established cognitive limits on working memory capacity: Miller's "The Magical Number Seven, Plus or Minus Two" [12] puts the number of items a person holds at once at about seven, and more recent work in cognitive modeling emphasises bounded processing capacity when interacting with complex environments [2]. The Qwen3-VL configuration treats length as a parameter to be measured rather than a constant to justify, and Section 6.3 reports accuracy across all eight lengths. The captioner overshoots the instruction consistently: a request for 3 words returns 3.7 on average and a request for 30 returns 34.3, with the standard deviation between the five repetitions of one image rising from 0.56 to 3.02 words. Table 7 reports the spread across all captions at each length, which is the larger of the two quantities.

Captions frequently name the class of the image they describe. In the Qwen3-VL configuration, 87% of captions contain their own class name, a synonym, or its head noun. This makes classification easier and the resulting explanation partly circular, so we treat it as a quantity to measure rather than a caveat: Section 6.2 reports the rate per class and tier, and repeats the experiments on captions with those terms removed.

Table 1. The two configurations. The method is identical in both; only the models differ.

	GPT-4o configuration	**Qwen3-VL configuration**
Captioning	GPT-4o	Qwen3-VL-32B-Instruct
Embedding	text-embedding-ada-002 (1536-d)	Qwen3-Embedding-4B (2560-d)
Detection	GPT-4o	Qwen3-VL-32B-Instruct
Weights	closed	open
Captions per image	1, at seven words	5, at each of eight lengths

3.2 Embedding-based classifier

Following the SMER methodology, the words from textual descriptions generated by the LLM are then converted into word embeddings. These embeddings are then aggregated into a single vector by averaging. Let us denote a caption of image A as a sequence of words $w_1, \dots, w_N$. We aggregate embeddings of words in descriptions as follows: if $A = (w_1, \dots, w_N)$, then

$$\tilde{A} = \frac{1}{N} \sum_{j=1}^{N} \tilde{w_j}, \tag{1}$$

which we refer to as the *SAVe* (Sentence Averaged Embedding). Note that each word w_j, as well as the average embedding $\tilde{A}$, is represented as a fixed-length embedding vector.

The resulting SAVe vector serves as input for a logistic regression model trained to predict class probabilities.

Averaging is what makes the attribution in Section 4.5 exact, and it has two consequences for the representation: it is invariant to word order, and it splits multi-word expressions such as *hot pot* or *fire truck* into independent components.

The multimodal model is used solely to generate textual descriptions. A separate text-embedding model maps words to vectors: *text-embedding-ada-002* at 1536 dimensions in the GPT-4o configuration, and *Qwen3-Embedding-4B* at 2560 dimensions in the Qwen3-VL configuration.

3.3 Keyword-based concept identification

Word-level importance is computed using SMER and, for comparison, LIME. These methods quantify the contribution of each word to the predicted class.

3.4 Concept detection using LLM

The highest-ranked words are then used to guide bounding box generation in the original image.

The prompt used for object detection is the same for every class:

> *"Provide bounding box coordinates for the object from ([list of top three features]). Format: x_min, y_min, x_max, y_max with integers only."*

Fig. 2. Object detection prompt where *[list of top three features]* is dynamically generated.

In the Qwen3-VL configuration the model answers per word: asked about three words it typically returns three labelled boxes rather than one box for the object. We take the first, since the words enter the prompt in SMER rank order and the first box therefore corresponds to the highest-ranked word. Coordinates are returned on a fixed 0–1000 grid regardless of image size and rescaled to pixels. Boxes are requested only for images carrying ImageNet bounding-box annotations, so every box can be scored; Section 6.6 reports the results.

4 Example

This section illustrates the full pipeline using a single example, from caption generation to attribution and bounding box localization.

4.1 Textual description generation

The LLM generates the caption: *"Bee pollinating vibrant purple flowers"* for the image in Figure 3a:



(a) *Original image*



(b) *TEBOL box generated by Qwen3-VL-32B-Instruct (red) against the ImageNet annotation (green), IoU 0.93*

Fig. 3. Worked example, Qwen3-VL configuration. Captions and bounding boxes are generated by Qwen3-VL-32B-Instruct and words are embedded with Qwen3-Embedding-4B. The caption generated for this image is *"bee pollinating vibrant purple flowers"*, classified as *bee* with $P = 0.994$. The three highest-ranked words for this image (*bee*, *pollinating*, *flower*) were passed to the detection prompt, which returned the box shown in red. *Image source: ImageNet dataset [4].*

4.2 Embedding transformation

Each word is embedded independently and the embeddings are aggregated with SAVe. Words are embedded with *text-embedding-ada-002* at 1536 dimensions in the GPT-4o configuration, and with Qwen3-Embedding-4B at 2560 dimensions in the Qwen3-VL configuration. The example in Table 2 uses the Qwen3-VL configuration.

4.3 Embedding aggregation

Now, we aggregate these embeddings into a single vector using the average aggregation formula in Eq. 1.

The SAVe is:

$$\tilde{A} = [-0.0053,\ 0.0091,\ \ldots,\ 0.0022] \tag{2}$$

4.4 Input to Logistic Regression

The resulting vector $[-0.0053,\ 0.0091,\ \ldots,\ 0.0022]$ in Eq. 2 represents the aggregated embedding of the input description—capturing the semantic content of *"bee pollinating vibrant purple flowers"*—and serves as input to a logistic regression model trained to predict class probabilities from these semantic representations. The model returns $P(bee) = 0.994$.

4.5 Word importance

In this example, LIME assigns a score to each feature (word), representing the approximate weight or coefficient of that word in a sparse linear model fitted to a randomly sampled neighborhood around the instance being explained. These scores approximate the model's behavior within a sampled neighborhood of the instance [3, 23].

Table 2 reports LIME importance scores for individual words. Unlike SMER, these scores are derived from a local surrogate model and cannot be transparently aggregated into the original classifier's sentence-level prediction. For this reason, no aggregated LIME probability is reported in Table 2 (marked as N/A).

On the other hand, the SMER score represents the logit (i.e., the raw output before applying the sigmoid function) of the probability that a text containing only the given word would be classified into the positive class. Logits can be either positive or negative: a positive value means the word supports the positive class, while a negative value indicates it supports the negative class. A key advantage of SMER is that the overall prediction logit can be exactly decomposed into the average of the individual SMER word-level logits, reflecting the linear structure of the underlying model, as expressed in Eq. 6.

Because logistic regression is a linear model, the SMER word-level logits are additive and can be aggregated into the sentence-level prediction. Let the model parameters be represented by the weight vector β and the bias (intercept) term β_0. For each word embedding $\tilde{w}_j$, we first define the word-level contribution as

$$z_j = \beta^\top \tilde{w}_j. \tag{3}$$

The corresponding SMER word-level logit is then given by

$$\lambda(w_j) = \beta_0 + z_j. \tag{4}$$

The overall sentence-level logit can be obtained in two equivalent ways. First, by applying the model directly to the sentence-averaged embedding $\tilde{A}$ (defined in Eq. 1):

$$\lambda(A) = \beta_0 + \beta^\top \tilde{A}. \tag{5}$$

Alternatively, by averaging the individual SMER word-level logits:

$$\lambda(A) = \frac{1}{N} \sum_{j=1}^{N} \lambda(w_j). \tag{6}$$

Because the dot product and the averaging operation are linear, these two expressions are mathematically identical:

$$\beta_0 + \beta^\top \tilde{A} = \beta_0 + \frac{1}{N} \sum_{j=1}^{N} \beta^\top \tilde{w}_j. \tag{7}$$

Due to linearity of the dot product and averaging operation, the sentence-level logit equals the average of the word-level logits under the linear model. When the resulting sentence logit is converted into a probability using the sigmoid function, the obtained value is identical to the model's predicted probability for the full sentence. For the caption in Table 2, the mean of the five z_j values is 4.8214; adding $\beta_0 = 0.2429$ gives 5.0643, and $\sigma(5.0643) = 0.9937$, which is the probability the model assigns to the averaged vector.

Since SMER scores are computed directly from the logistic regression coefficients, they achieve perfect fidelity with the model being explained [6]. This is illustrated in Table 2: the mean of the word-level logits equals the sentence logit, and its sigmoid equals the model's predicted probability.

Table 2. Word importance for the caption *"bee pollinating vibrant purple flowers"* (image `n02206856_1815`, held out in fold 1), Qwen3-VL configuration. The logistic regression model outputs the probability of the *bee* class. $z_j = \beta^\top \tilde{w}_j$ is the word-level contribution and $\lambda(w_j) = \beta_0 + z_j$ the SMER word logit; the intercept β_0 is a single fitted constant, repeated down the column. The final row is the averaged caption vector: its z is the mean of the column above it, so $\text{mean}(z_j) + \beta_0$ reproduces the model's sentence logit exactly (Eq. 6). LIME scores come from a local surrogate and cannot be aggregated into the classifier's prediction, so no aggregate is reported.

Input data		**SMER**					**LIME**
Word	**Embedding**	$\mathbf{z_j}$	β_0	**Logit** $\lambda(w_j)$	**P(bee)**	**P(ant)**	**Importance**
bee	[-0.0003, -0.0080, ..., -0.0033]	13.4518	0.2429	13.6946	> 0.9999	< 0.0001	0.1224
pollinating	[0.0115, -0.0303, ..., 0.0002]	6.9856	0.2429	7.2285	0.9993	0.0007	0.0854
vibrant	[0.0040, 0.0526, ..., -0.0138]	-1.1084	0.2429	-0.8655	0.2962	0.7038	-0.0124
purple	[-0.0147, 0.0403, ..., 0.0308]	-0.5149	0.2429	-0.2721	0.4324	0.5676	-0.0083
flowers	[-0.0267, -0.0090, ..., -0.0028]	5.2931	0.2429	5.5360	0.9961	0.0039	0.0689
Results on averaged word vectors (SAVe(caption))							
SAVe(caption)	[-0.0053, 0.0091, ..., 0.0022]	4.8214	0.2429	5.0643	0.9937	0.0063	N/A

4.6 Object detection

The three highest-ranked words for this image across its five captions (*bee*, *pollinating*, *flower*) are inserted into the detection prompt (Fig. 2). The returned box, shown in red in Fig. 3b, overlaps the ImageNet annotation with an intersection-over-union of 0.93.

5 Experimental setup

This section describes the images used, the textual versions derived from them, and the protocol under which every result is obtained. Both configurations are evaluated on the same images and the same tasks; they differ only in the models used and, consequently, in the evaluation protocol. The evaluation itself follows in Section 6.

5.1 Dataset

We used a subset of the *ImageNet* dataset [4] comprising ten object classes, organised as five binary classification tasks. In total 14,351 images were used; Table 3 gives the per-task counts. By separating the dataset into these five binary problems, we aimed to compare model performance and interpretability across different object categories.

Table 3. Classification tasks and image counts. Synsets are disjoint across tasks. Both configurations use cross-validation over all images of a task; the Train and Test columns give the sizes of the 80/20 split within one fold.

Task	Synsets	Images	Train	Test
acoustic guitar / violin	n02676566, n04536866	3,347	2,678	669
ambulance / fire truck	n02701002, n03345487	2,866	2,293	573
ant / bee	n02219486, n02206856	3,328	2,662	666
cucumber / zucchini	n07718472, n07716358	2,361	1,889	472
hot pot / vase	n07590611, n04522168	2,449	1,959	490
Total		**14,351**	**11,481**	**2,870**

Two further annotation sources are used. 4,531 images carry ImageNet bounding-box annotations [4], which provide the ground truth for the localization evaluation in Section 6.6. 4,725 images are covered by ImageNet-Captions [8], which recovers user-supplied metadata for ImageNet images; we use its tag field as a human-authored point of comparison for the generated descriptions.

5.2 Data preparation and dataset versions

To obtain consistent semantic representations, we generated descriptions for all images with fixed prompting, in both configurations. No fine-tuning was applied. Although ImageNet provides class labels, many entries lack natural language descriptions. ImageNet-Captions [8] supplies uploader metadata instead. For instance, consider the ImageNet-Captions record for a vase image shown in Figure 4, which includes tags supplied by the uploader:


```
"filename": "n04522168_30221.JPEG", "title": "IMG_4920 Tokyo National
Museum", "tags": ["Tokyo National Museum", "Tokyo", "Japan", "Ueno", "vase",
"jar", "rooster", "floral"], "description": "", "wnid": "n04522168"
```


Fig. 4. Example of an ImageNet-Captions record for a vase image.

The tags name the object and its setting but do not describe its appearance. They are nonetheless a human-authored representation of the same image, and we use them in Section 6.2 as a point of comparison for the generated descriptions.

Generation of LLM-based descriptions.

In the GPT-4o configuration, captions were generated with OpenAI's *gpt-4o* via the *chat.completions.create* endpoint, at *temperature* = 1.0, with the image supplied alongside the prompt of Section 3.1 in a multimodal request. In the Qwen3-VL configuration, Qwen3-VL-32B-Instruct was served locally at the same temperature, with the reply length budget scaled to the requested caption length and a seed recorded for each repetition. In that configuration each image was described at eight lengths with five repetitions, giving 574,040 generated descriptions.

Dataset versions.

Each of the five tasks is evaluated under three versions of its textual input:

- *captions* — the descriptions as generated;
- *captions without class terms* — the same descriptions with the class name, its agreed synonyms and its head noun removed (Section 6.2);
- *tags* — the ImageNet-Captions tags for the same images.

Because tags cover only part of the corpus, comparisons against them are also run on the subset of images that carry tags, with the description versions restricted to those same images, so that the comparison is not confounded by which photographs each version contains.

5.3 Configurations

Both configurations use 5-fold cross-validation repeated five times, giving 25 fits per condition on the same folds; the spread across those fits is what makes confidence intervals and significance tests possible. Folds are assigned to images rather than to descriptions, so the five descriptions of one photograph in the Qwen3-VL configuration cannot be split between training and testing. The GPT-4o configuration is evaluated from the descriptions and the `text-embedding-ada-002` word vectors stored from its original run, and covers four of the five tasks: for cucumber / zucchini the stored word vectors are 300-dimensional rather than ada-002, so that task is not rerun.

Where an image has several descriptions, their predicted probabilities are averaged into one probability per image before thresholding. This scores the photograph rather than the sentence, and averaging rather than voting retains how close the decision was.

All conditions reuse the same folds, so differences between them are paired. Because repeated cross-validation produces correlated fits, a standard paired t-test would overstate significance; we use the Nadeau–Bengio corrected resampled t-test with Bonferroni correction within each comparison family.

5.4 Quantitative evaluation

All the experiments are reported in Section 6; this section states what each compares and how it is measured. Table 4 summarises them.

Table 4. Summary of the experimental design.

Experiment	Config.	Comparison	Metric
Classification performance	both	the pipeline against seven alternative methods	image-level accuracy; macro precision, recall, F1, ROC-AUC
Label leakage	Qwen3-VL	descriptions with vs. without class terms	incidence rate; paired difference in accuracy
Caption length	Qwen3-VL	eight lengths, 3 to 30 words, two textual conditions	accuracy per length
Attribution faithfulness	both	SMER against LIME	area over the perturbation curve (AOPC)
Attribution stability	Qwen3-VL	refitting, regeneration, reseeding	top-3 overlap; top-1 change rate; Kendall's τ
Localization	Qwen3-VL	predicted boxes against ImageNet annotations	IoU; success rate; box drift across repetitions
Multi-class extension	Qwen3-VL	three classes, one-vs-rest	accuracy; logit reconstruction error

Classification performance. In the Qwen3-VL configuration we evaluate, on the same images and folds, the classifier trained on the generated descriptions, on descriptions with class terms removed, and on the ImageNet-Captions tags; string matching for either class name with no model; the captioning model asked to classify the image in a single call; CLIP and CoCa choosing the nearer of two class prompts; and logistic regression on frozen CLIP and CoCa image features. In the GPT-4o configuration we evaluate the classifier on seven-word descriptions and logistic regression on frozen VGG16 features, on the same folds. We report image-level accuracy with macro-averaged precision, recall, F1 and ROC-AUC.

Label leakage. Class terms come from a list fixed in advance: 105 terms in three tiers — the class name and its morphological variants (39), agreed synonyms (59), and the standalone head noun of a two-word class name (7). We report the incidence of class terms by class and tier, and repeat the classification above on descriptions with those terms removed and re-embedded. The terms of both classes of a task are removed from every description, so the removal does not use the image's label; removal is repeated until no term remains, since deleting a word can join its neighbours into a term. Because the naming rate itself varies with length, descriptions that name their class are additionally compared against those that do not within each length, holding length fixed.

Caption length. The initial seven-word constraint followed Miller's account of working memory [2, 12] and was never measured against classification. In the Qwen3-VL configuration, descriptions were therefore generated at eight lengths — 3, 5, 7, 10, 15, 20, 25 and 30 words — with five repetitions at each, and the classifier refitted independently at every length, both for descriptions as generated and for descriptions with class terms removed. The GPT-4o configuration has a single length and does not enter this comparison. The same partitions are scored at every length, so the curves are paired. Accuracy is reported at every length for each of the two textual conditions.

Attribution faithfulness. SMER is evaluated against LIME as the reference method, using the area over the perturbation curve (AOPC) [7]. For $k = 0, 1, \ldots$, the k words ranked highest by each method are ablated from a description and

the decrease in the predicted probability of the selected class is recorded; the resulting curve is integrated by the trapezoidal rule to give the AOPC score. Ablation is capped at $n - 1$ words, and a description exhausted earlier retains its final value, so all descriptions contribute at every k. LIME is configured to treat each word position as a distinct feature, so that both methods rank the same objects, and is evaluated on a seeded sample stratified by fold and class, with the SMER curves recomputed on that sample. The Qwen3-VL results are reported in Section 6.4; the GPT-4o configuration is reported in Appendix A.

Attribution stability. Three sources of randomness are varied independently: refitting the classifier across the 25 splits, regenerating the description five times at temperature 1.0, and reseeding LIME five times. SMER admits no reseeding axis, as its score involves no sampling. For each source we report the mean pairwise Jaccard overlap of the three highest-ranked words, the proportion of runs whose highest-ranked word differs from the modal choice, and Kendall's τ over the full ordering where the same words are ranked in every run.

Localization. Bounding boxes are evaluated against ImageNet's own annotations on the 4,531 images that carry them, in the Qwen3-VL configuration at five-word descriptions. The three highest-ranked words for an image are passed to the detection prompt and the box returned for the highest-ranked word is scored. Each image is queried five times. We report the mean intersection over union with the annotated box, the proportion of boxes reaching an intersection over union of 0.7, the movement of the box edges across repetitions, and the rate at which a word returns no coordinates at all.

Multi-class extension. The decomposition requires only that pooling be a mean and the logit be linear in it, so it extends to any number of classes. We verify this on cucumber, zucchini and hot pot, fitted one-vs-rest rather than multinomially, since a softmax fit is identified only up to a shift and an absolute per-word contribution would carry no meaning. We report classification accuracy and the maximum absolute difference between the model's logit and the logit reconstructed from its per-word contributions.

6 Results

This section reports the evaluations described in Section 5.4, in the same order. Results in the Qwen3-VL configuration are at five-word descriptions unless stated otherwise; results in the GPT-4o configuration are at seven words.

6.1 Classification performance

Table 5 reports accuracy for the TEBOL pipeline and the other baselines.

Table 5. Image-level accuracy. The block headings denote the configuration of the pipeline and the evaluation protocol, not the model used by every row: CLIP and CoCa are independent of both captioners. In the Qwen3-VL configuration descriptions are five words, and each value is the mean over 25 fits—5-fold cross-validation repeated five times, with folds assigned to images; ± is the 95% half-width, a t-interval over those fits where the classifier is refitted and a bootstrap over images from 5,000 resamples where it is not, so the two kinds are not comparable as widths. The GPT-4o configuration uses seven-word descriptions, one per image, on the same folds and with the same intervals; it omits cucumber / zucchini, whose stored word vectors are not ada-002. VGG16 is a logistic regression on frozen features with torchvision's ImageNet weights, fitted on the same folds. The blocks differ in captioner, embedding and the number of descriptions per image. Bold marks the highest value in each column within a configuration; where intervals overlap, the difference is not resolvable.

	guitar/ violin	**ambul./ truck**	**ant/ bee**	**cucum./ zucch.**	**hotpot/ vase**
Qwen3-VL configuration					
LR on SAVe (TEBOL)	**0.9907** ±.001	0.9852 ±.002	**0.9849** ±.002	**0.9410** ±.004	0.9976 ±.001
Class-name matching	0.8440 ±.012	0.9225 ±.010	0.8975 ±.010	0.8191 ±.015	0.6460 ±.019
Zero-shot classification	0.9848 ±.004	0.9853 ±.004	0.9772 ±.005	0.8882 ±.013	0.9980 ±.002
CLIP	0.9737 ±.005	0.9822 ±.005	0.9426 ±.008	0.6709 ±.019	0.9951 ±.003
LR on CLIP features	0.9822 ±.002	**0.9861** ±.002	0.9689 ±.003	0.8242 ±.005	**0.9992** ±.001
CoCa	0.9788 ±.005	0.9787 ±.005	0.9384 ±.008	0.7150 ±.018	0.9980 ±.002
LR on CoCa features	0.9830 ±.002	0.9788 ±.002	0.9606 ±.003	0.8240 ±.005	0.9988 ±.001
GPT-4o configuration					
LR on SAVe (TEBOL)	**0.9611** ±.003	**0.9691** ±.003	0.9449 ±.004	—	0.9931 ±.001
LR on VGG16 features	0.9537 ±.002	0.9435 ±.004	**0.9456** ±.004	**0.8138** ±.008	**0.9987** ±.001

In the Qwen3-VL configuration the TEBOL (LR on SAVe) is the highest-scoring method on three of the five tasks, leading every alternative there by 0.008 to 0.270. The remaining two sit near ceiling, where no method separates from the rest. Class-name matching is lowest throughout and falls to 0.646 on hot pot / vase, the pair whose descriptions name their class least often. Zero-shot classification is the closest alternative, within 0.008 of the pipeline on four tasks but 0.053 lower on cucumber / zucchini, where CLIP and CoCa fall further still, to 0.671 and 0.715; fitting a classifier on their frozen features recovers part of that gap but leaves 0.117.

In the GPT-4o configuration the pipeline exceeds the VGG16-feature classifier on ambulance / fire truck (0.9691 against 0.9435, corrected resampled t-test $p < 0.001$), is level with it on acoustic guitar / violin and on ant / bee ($p = 0.11$ and $p = 0.91$), and trails it on hot pot / vase (0.9931 against 0.9987, $p = 0.005$). Cucumber / zucchini is not rerun in this configuration; the VGG16 classifier reaches 0.8138 there.

6.2 Label leakage

Table 6 reports accuracy for descriptions as generated, for the same descriptions with class terms removed, and for the human tags, all on the images ImageNet-Captions covers.

Table 6. Label leakage at five-word descriptions, Qwen3-VL configuration. All three conditions are evaluated on the images ImageNet-Captions covers, using the same fold assignment. Each is the mean over 25 fits with the 95% half-width. Bold marks the highest value in each column.

	guitar/ violin	ambul./ truck	ant/ bee	cucum./ zucch.	hotpot/ vase
Descriptions (TEBOL)	**0.9855** ±.004	0.9924 ±.002	**0.9852** ±.003	**0.9277** ±.008	**0.9979** ±.002
class terms removed	0.7585 ±.011	0.8185 ±.012	0.8627 ±.006	0.6891 ±.009	**0.9979** ±.002
Tags	0.9770 ±.004	**0.9929** ±.003	0.9578 ±.005	0.9240 ±.006	0.9709 ±.005

Descriptions name the class of the image they describe in 65% to 93% of cases, depending on the task. Removing those terms costs between 0.123 and 0.239 accuracy on four of the five tasks, and nothing on hot pot / vase, where the two conditions agree to four decimal places: food scenes and decor are separable without being named, the other four pairs are not. Splitting descriptions by whether they happen to contain a class term rather than removing it gives the same picture over all images — those that do are classified at 0.997 to 0.999 in every task, those that do not at 0.710 to 0.993.

The tier structure of the term list matters for the two-word classes. Acoustic guitar is named in full in 69.9% of its descriptions but contains the head noun *guitar* in 98.3%; hot pot is named in full in only 34.6% but contains *pot* in 46.4%.

6.3 Caption length

Table 7 reports how the captioner responds to the length instruction, and Figure 5 reports accuracy at each length for both textual conditions.

Table 7. Caption generation, Qwen3-VL configuration, pooled over the five tasks. Each row covers the 71,755 captions produced for one value of n in the prompt *"Describe this image in n words"* — one for each of the 14,351 images at each of five repetitions. *Names its class* is the share of captions containing the class name, an agreed synonym or its head noun; *non-Latin* is the share of captions containing Chinese or Japanese characters.

Words requested	Words produced (mean ± SD)	Names its class	Non-Latin
3	3.71 ± 1.25	79.0%	0.02%
5	5.70 ± 0.91	85.8%	0.01%
7	7.22 ± 1.13	86.9%	0.02%
10	10.30 ± 1.55	88.3%	0.04%
15	16.76 ± 2.40	88.8%	0.13%
20	22.73 ± 3.18	89.1%	0.19%
25	27.31 ± 3.92	89.3%	0.24%
30	34.39 ± 4.08	90.0%	0.30%

The requested length is followed only approximately. The captioner overshoots at every setting, producing 3.71 words when three are asked for and 34.39 when thirty are, and the spread across repetitions widens from 1.25 to 4.08 tokens. Two further properties change with length: the share of captions naming their own class rises from 79.0% to

90.0%, and the share containing Chinese or Japanese characters rises from 0.02% to 0.30%, the latter concentrated on hot pot and on ceramics.

Accuracy depends on caption length, and the dependence is not monotone (Figure 5), falling 0.083 on cucumber / zucchini and 0.048 on ant / bee between five and thirty words; hot pot / vase is flat, varying by 0.003 across the whole range. Five words is therefore the length at which results are reported in the paper.

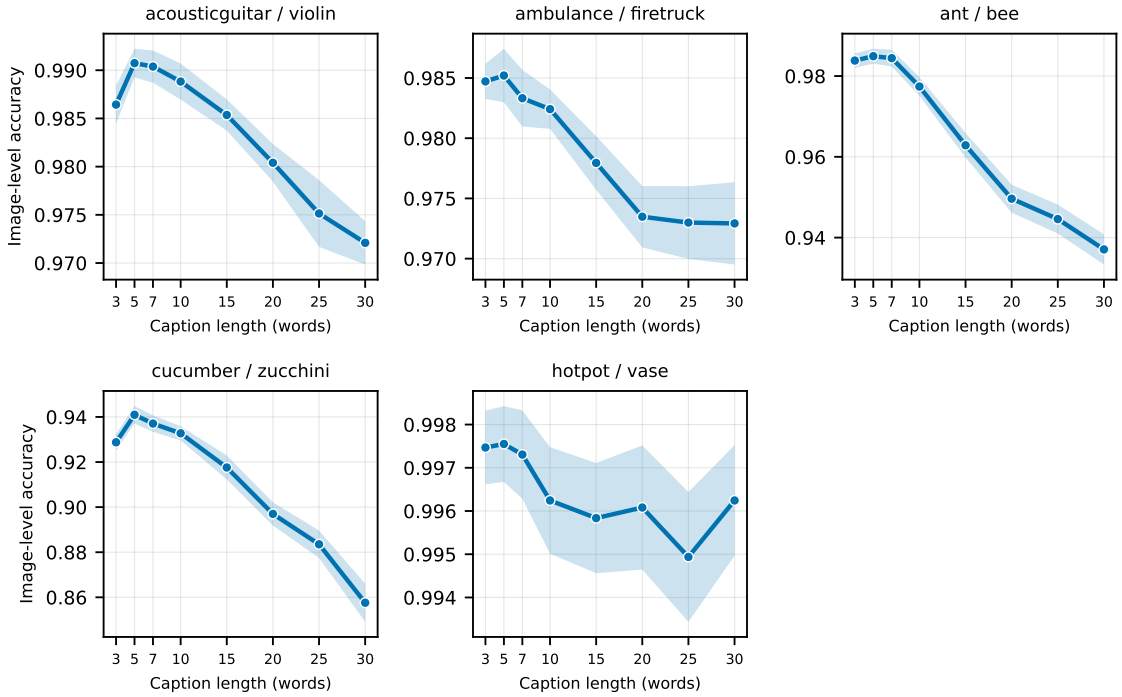


Fig. 5. Image-level accuracy against caption length for descriptions as generated, Qwen3-VL configuration. Bands are 95% t-intervals over the 25 fits. Accuracy peaks at five words on all five tasks and declines thereafter on four of them.

6.4 Attribution faithfulness

To evaluate the identification of key concepts, we use the AOPC (Area Over the Perturbation Curve) metric. AOPC quantifies how much the model's confidence decreases when the features an explanation ranks highest are removed. It is computed by progressively removing the top-ranked features and recording the drop in the predicted probability of the original class [7]. A steeper curve means the ranking identified features the model relied on more heavily. The curves in Figures 6 and 7 come from the Qwen3-VL configuration at five words, averaged over the 25 fits of the repeated cross-validation, with 95% t-intervals as bands.

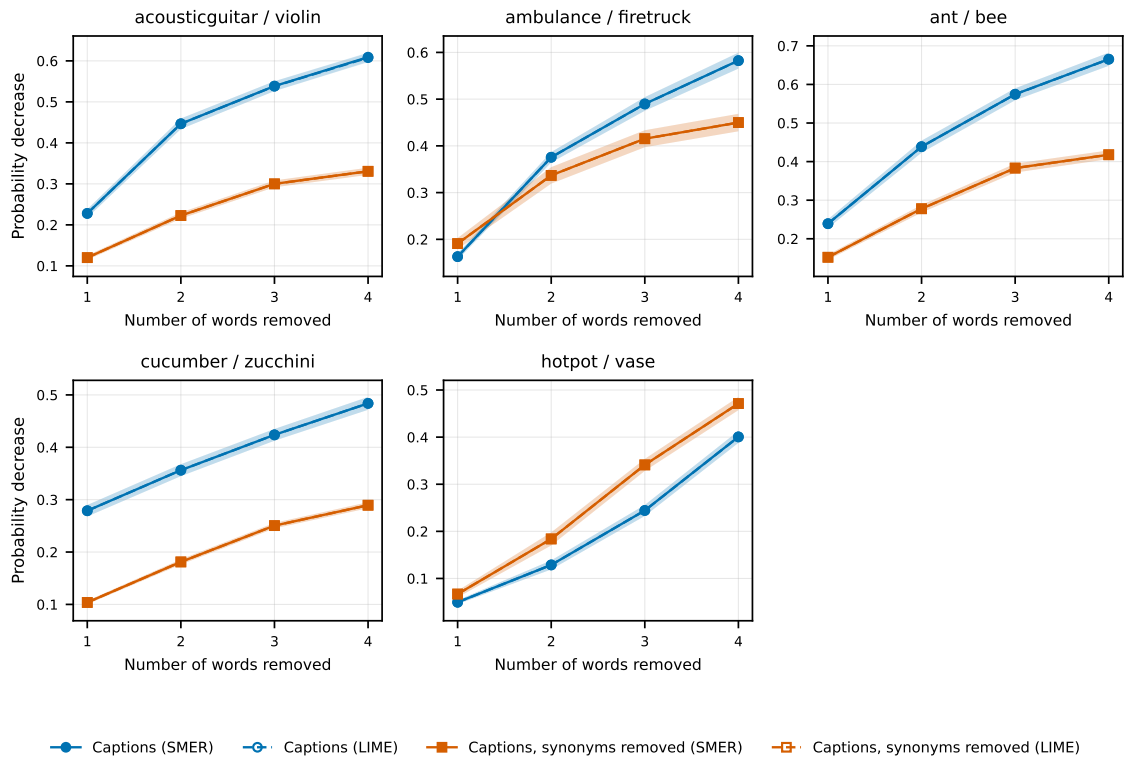


Fig. 6. Perturbation curves for descriptions as generated and with the class name and its synonyms removed, five words, Qwen3-VL configuration. Solid lines with filled markers are SMER, dashed lines with hollow markers LIME, one colour per condition. LIME coincides with SMER on every task, so its curves lie behind SMER's. Bands are 95% t-intervals over the 25 fits.

SMER and LIME give the same curve on all five tasks in both conditions. On descriptions as generated their integrated scores differ by at most 0.001 (paired t-test over tasks $p = 0.116$); with the class name and its synonyms removed they differ by less than 0.001 ($p = 0.096$).

Removing the class terms lowers the curve on four tasks. Hot pot / vase is the exception, rising from 0.623 to 0.828: once the class term is gone the remaining words carry proportionally more of the decision, so each removal costs more. This is the task whose descriptions name their own class least often, and the one whose classifier starts from the lowest confidence.

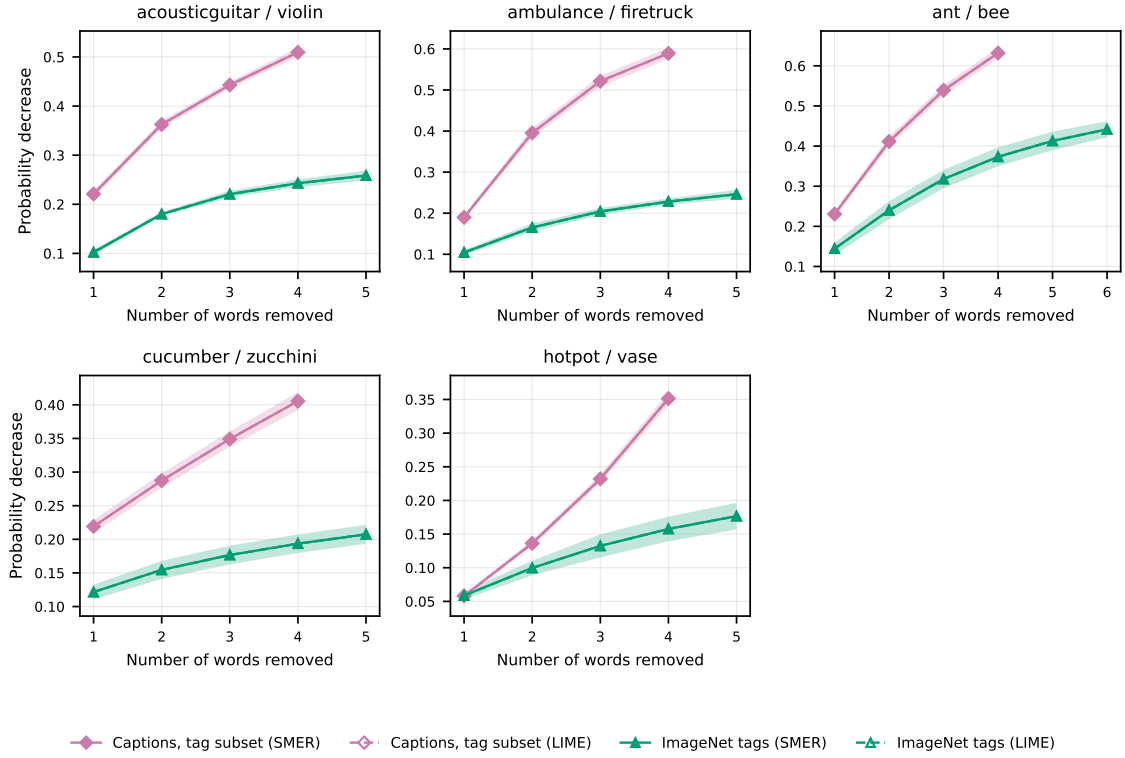


Fig. 7. Perturbation curves for ImageNet-Captions tags against descriptions of the same images, Qwen3-VL configuration; line styles as in Figure 6. Tags exist for 32.9% of the images and both conditions are restricted to those, so the comparison separates representation from sample. A tag list holds 8.2 tags on average, between 7.4 and 9.3 by task, against 5.7 word tokens in a five-word description; since removal is capped at $n - 1$, the tag curves extend one to two steps further, and ant / bee, which carries the most tags, reaches $k = 6$. LIME coincides with SMER on every task, so its curves lie behind SMER's.

Descriptions support the more faithful ranking than tags on four of the five tasks, reaching 0.602 to 1.402 where tags on the same images reach 0.537 to 0.875. Ant / bee is the single reversal, tags reaching 1.709 against 1.497; it is also the task with the most tags per image.

SMER and LIME coincide on tags as well as on descriptions: their integrated scores differ by less than 0.001 on every task, for tags and for descriptions of the same images.

6.5 Attribution stability

An explanation that changes when nothing of substance changes is not a description of the model. Three sources of randomness are varied independently, holding the other two fixed: refitting the classifier across the 25 cross-validation splits, regenerating the description five times at temperature 1.0, and reseeding LIME five times. SMER admits no reseeding axis, as its score involves no sampling.

Three measures are reported, because agreement at one resolution does not imply agreement at another. The Jaccard overlap of the three highest-ranked words asks whether the words a reader sees stay the same, and a set measure suits them because their order among themselves is not shown. The proportion of runs whose highest-ranked word differs

Table 8. Attribution stability, descriptions as generated at five words, Qwen3-VL configuration. Both sources hold the description fixed, so the same words are ranked in every run and the explainer is the only thing varying. Values are the mean ± standard deviation over the five tasks. Bold marks the better value within each block; under refitting the two methods differ by less than the deviation across tasks. SMER's reseeding values carry no deviation because they do not vary: $z = \beta \cdot e$ involves no sampling, so they follow from the definition rather than from measurement.

Source of randomness	Method	Jaccard@3	Top-1 changed	τ
Refitted classifier	SMER	**0.961** ± 0.034	0.003 ± 0.002	**0.881** ± 0.087
	LIME	0.955 ± 0.034	0.003 ± 0.002	0.876 ± 0.082
Reseeded explainer	SMER	**1.000**	**0.000**	**1.000**
	LIME	0.977 ± 0.015	0.001 ± 0.001	0.934 ± 0.048

from the modal choice asks the same question at the sharpest point, since the leading word can flip while the top three remain intact. Kendall's τ covers the full ordering, on which the perturbation curve of Section 6.4 depends at every removal step. Jaccard and τ rise with stability, the top-1 figure falls with it. All three record consistency, not correctness: a reproducibly wrong ranking would score at the ceiling on each.

Under refitting the two methods are indistinguishable: top-three overlap 0.961 against 0.955, τ 0.881 against 0.876, and the leading word changing in 0.3% of runs for both. This is the only axis on which SMER can move, since it is exact given a model rather than given a dataset. The distance between the overlap and τ figures is why both are reported: the three leading words survive refitting almost unchanged while the tail of the ordering moves considerably more.

Reseeding separates the two methods by construction. LIME retains 0.977 of its top-three overlap and $\tau = 0.934$ across five seeds; SMER returns the identical ranking on every seed, having no seed to vary.

Regenerating the description moves the ranking further than either, and that variation belongs to the captioner. Five descriptions of the same photograph share 0.640 of their words; the three words SMER ranks highest across those descriptions share 0.663. The attributed words are therefore no less stable than the text they are drawn from. Only 19 to 28% of images receive five identical descriptions, yet 94.6 to 99.6% receive the same predicted class from all five, so the captioner's variation reaches which words are named before it reaches what is predicted.

Appendix B gives the per-task values and repeats the measurement at seven and thirty words.

6.6 Localization

The words SMER ranks highest should point at the object the classifier used. Bounding boxes are evaluated against ImageNet's annotations on the 4,531 images that carry them, in the Qwen3-VL configuration at five-word descriptions. The three highest-ranked words for an image are passed to the detection prompt and the box returned for the highest-ranked word is scored. Each image is queried five times at non-zero temperature, giving 22,655 calls.

The highest-ranked word localizes the annotated object with a mean intersection over union of 0.829, and 80.6% of boxes reach 0.7. Performance tracks how bounded the object is. Ambulance / firetruck is highest at 0.907, both classes being large and centrally framed; hot pot is the lowest class at 0.697, a table scene rather than a bounded object, so annotation and model disagree about its extent. Acoustic guitar reaches 0.936 against violin's 0.711, the latter photographed at more varied scales and angles. Boxes are reproducible: edges move 1.8% of the image dimension across the five calls, and 14 of 22,655 calls fail to parse.

Not every ranked word yields a box. Among the 21% of replies that name their boxes, 14,085 occurrences of a word in an image's top three can be checked; 647 of them returned no coordinates, a rate of 4.6%, of which nine were explicit

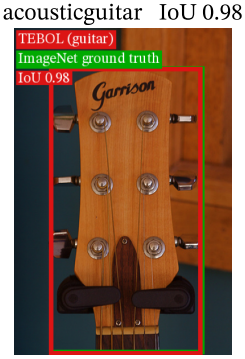
Table 9. Localization of the highest-ranked word against ImageNet annotations, Qwen3-VL configuration, five-word descriptions, five repetitions per image. IoU is the mean over all calls ± the half-width of its 95% confidence interval; IoU ≥ 0.7 is the share of calls clearing that threshold; drift is the mean movement of the box edges across the five repetitions, as a percentage of the image dimension.

Task	Images	Mean IoU	IoU ≥ 0.7	Drift
acousticguitar / violin	942	0.833 ± 0.015	77.5%	1.8%
ambulance / firetruck	936	0.907 ± 0.010	92.6%	1.4%
ant / bee	915	0.846 ± 0.012	86.6%	1.0%
cucumber / zucchini	698	0.758 ± 0.023	73.7%	2.5%
hotpot / vase	1040	0.788 ± 0.014	72.0%	2.3%
All	4,531	0.829 ± 0.007	80.6%	1.8%

refusals. A further 1.7% of calls returned no box for the highest-ranked word and are scored as failures. The words that fail name no bounded object: *fire* returns no box in 43.1% of its appearances for firetruck images and *guitar* in 23.5% for acoustic guitar. This bounds what the evaluation can show, since a faithful word naming a texture, a colour or an activity has nothing to box.

Figure 8 shows one example per class.

acousticguitar IoU 0.98



"Garrison guitar headstock with tuning pegs"

ambulance IoU 0.91



"White ambulance with red and green stripes"

ant IoU 0.74



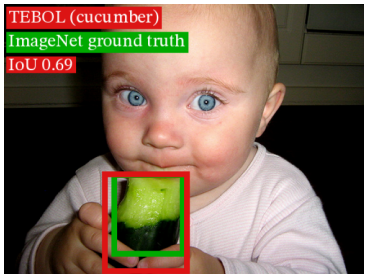
"Ant, eggs, leaf, green, nature"

bee IoU 0.89



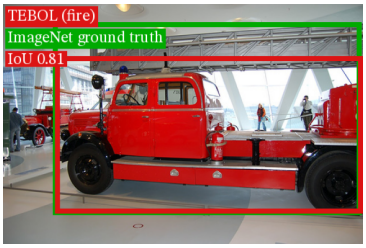
"Bumblebee on purple lavender bloom"

cucumber IoU 0.69



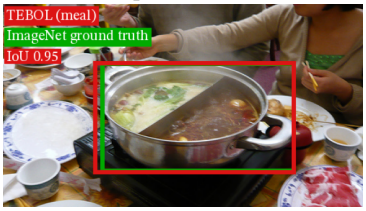
"Baby with blue eyes eating cucumber"

firetruck IoU 0.81



"Red vintage fire truck on display"

hotpot IoU 0.95



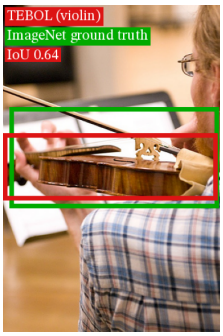
"Hot pot dining with friends"

vase IoU 0.97



"Rusty, old ceramic jug with handles"

violin IoU 0.64



"Man playing violin with sheet music"

zucchini IoU 0.16



"Zucchini salad with olives and cheese"

Fig. 8. Localization examples, one per class, Qwen3-VL configuration at five words. Green is the ImageNet annotation, red the box returned for the highest-ranked word, which is named in the corner together with the IoU. Disagreements are mostly about extent rather than identity: for zucchini the annotation covers the whole dish while the box encloses one slice, and for violin the box takes the instrument's body where the annotation includes the neck.

6.7 Multi-class extension

The decomposition requires only that pooling be a mean and the logit be linear in it, so it extends to any number of classes. We verify this on cucumber, zucchini and hot pot in the Qwen3-VL configuration at five-word descriptions, fitted one-vs-rest rather than multinomially: a softmax fit is identified only up to a shift of all logits, so an absolute per-word contribution would carry no meaning, while each one-vs-rest classifier is the same binary problem the preceding sections validate. The three classes give 3,510 images and 17,550 descriptions, evaluated under the same 5-fold × 5-repeat protocol.

Image-level accuracy is 0.950 ± 0.003 against a chance rate of 0.333, with balanced accuracy 0.951 and macro F_1 0.950. The errors fall where the classes resemble one another: hot pot, a prepared dish, is recovered on 99.6% of its images, while cucumber and zucchini are confused with each other on 6.0% and 5.5% respectively and almost never with hot pot.

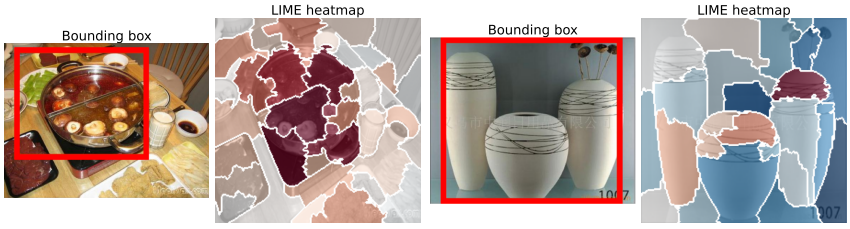
The decomposition holds exactly. Reconstructing each description's logit as its bias plus the mean of its words' per-class contributions, and comparing against the model's own logit over 1,600 descriptions in all three classes, the largest absolute difference is 5.01×10^{-6} — floating-point error, not approximation. The highest-weighted words are the class terms and their morphological variants: *cucumber* and *cucumbers* for cucumber, *zucchini* and *zucchinis* for zucchini, and *stew*, *stewing* and *hotpot* for hot pot.

7 Qualitative evaluation and user study

While the previously presented evaluation focused on numerical assessment, in this section, we compare TEBOL with LIME visually. The visualizations in this section and the user study that follows were produced with the GPT-4o configuration at seven-word descriptions.

To improve the objectivity of the comparisons, both methods were combined with logistic regression as the underlying model. However, TEBOL is applied to text-based explanations, while LIME operates on features extracted from image content via VGG16.

Figure 9 shows the comparison for hot pot and vase; the remaining four class pairs are in Appendix C. LIME explains the class predicted by a logistic regression on frozen VGG16 features, the classifier of the *LR on VGG16 features* row of Table 5, and each superpixel is tinted by the weight LIME fitted for it: dark red raises the probability of the predicted class and blue lowers it. In the hot pot example the largest weights lie on the pot. In the vase example they lie on the top of the right vase, while the bodies of the vases carry negative weights. LIME's superpixels are cut from colour and texture and do not follow the outline of an object, so one object is covered by several superpixels whose weights differ, and in the appendix examples of the fire truck, the ant and the zucchini the largest positive weights lie beside the object. The TEBOL box is drawn around the object that the highest-ranked word names. Section 6.6 evaluates those boxes against the ImageNet annotations; the comparison with LIME here is qualitative, and the user study of Section 7.3 compares the boxes with Grad-CAM heatmaps.



(a) *"Hot pot with various ingredients and condiments."* (b) *"Three decorative vases with minimalistic designs."*

Fig. 9. Examples of a bounding-box explanation generated by our TEBOL method (left) and a LIME explanation (right) for Vase vs. Hotpot classification. LIME explains the class predicted by a logistic regression on VGG16 features: each superpixel is tinted by its LIME weight, dark red raising and blue lowering the probability of that class. *Images source: ImageNet dataset [4].*

7.1 Evaluation of visualizations

In addition to the quantitative and qualitative comparisons, we conducted a user study to evaluate the interpretability of different explanation formats. The goal of this human evaluation was to assess how intuitively people understand two popular types of computer vision explanations: bounding boxes and feature importance heatmaps.

At the same time, these visualization formats should not be interpreted as exposing the same explanatory signal. Grad-CAM visualizes spatial regions used by a ResNet classifier, whereas TEBOL first identifies important words through SMER and then asks an external multimodal LLM to localize the corresponding concepts. Consequently, TEBOL's bounding boxes represent post-hoc semantic grounding of faithful textual attributions, not a direct visualization of the classifier's internal spatial attention. The comparison in this section therefore concerns human interpretability and communicative usefulness of the resulting visualizations rather than strict equivalence of model-internal evidence.

There were two possible methods for generating heatmaps: Grad-CAM and LIME. However, the heatmaps produced by these methods differ significantly in appearance and behavior. Therefore, as a first step, we performed a qualitative comparison between Grad-CAM and LIME. Based on this comparison, we selected Grad-CAM as the reference method for the user study.

The remainder of this section is divided into two parts: the first subsection presents the comparison between Grad-CAM and LIME, and the second subsection describes the design and results of the user study.

7.2 LIME vs. Grad-CAM Using a ResNet Classifier

In addition to the explanation techniques LIME and SMER, we explored the *Grad-CAM (Gradient-weighted Class Activation Mapping)* method to further evaluate the interpretability of our image classification models.

To ensure a fair comparison between LIME and Grad-CAM, we applied both techniques to the same underlying model—a ResNet classifier [24]. This model was selected because Grad-CAM is specifically designed to work with deep neural networks. Using a consistent base model allowed us to directly compare how each explanation method attributes importance to different image regions [5].

This ResNet-based comparison also clarifies the role of Grad-CAM in the article. We use Grad-CAM as the reference heatmap format for the user study because its map is a single continuous region over the object, where LIME's is a set of superpixel weights, not because it is theoretically equivalent to TEBOL's grounding mechanism. TEBOL's

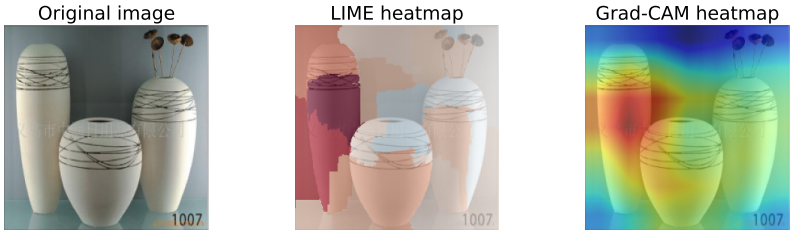
boxes answer the question “where is the SMER-selected concept located according to the external multimodal LLM?” whereas Grad-CAM answers “which image regions most influenced the ResNet prediction?” These are related but distinct explanatory targets.

By applying both methods to the ResNet model, we were able to directly compare the quality and consistency of the explanations provided by each technique. This experimental setup enabled us to evaluate how effectively each method identified the most important regions in the image, and whether their highlighted areas aligned with features critical for classification.

The results, shown in Figure 10, are two different kinds of map for the same class of the same network. Grad-CAM gives a smooth map over the object: on the hot pot the strongest activation covers the pot, and on the vases it is highest on the left vase. LIME gives one weight per superpixel. On the hot pot its strongest weights lie on the pot, and on the vases on the left vase, but the superpixels are cut from colour and texture and do not follow the outline of the object, so one object carries several weights. Grad-CAM draws the explanation as one continuous region over the object, and LIME as a set of superpixel weights.



(a) *”Hot pot with various ingredients and condiments.”*



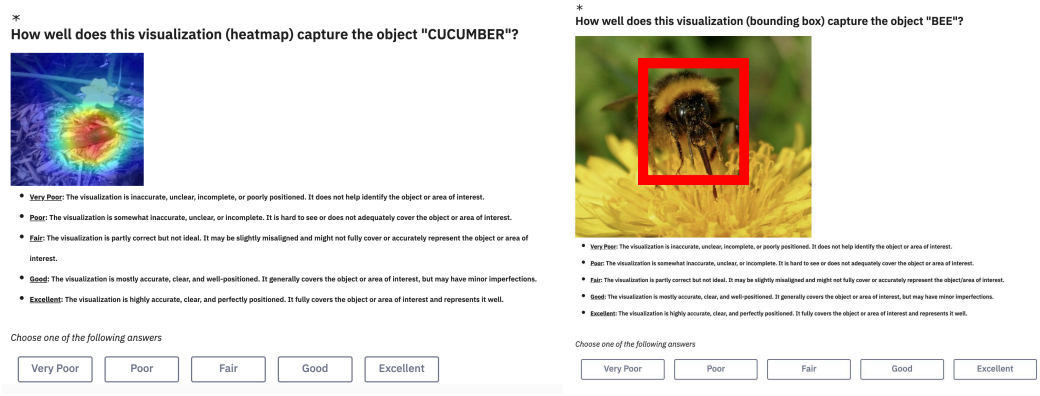
(b) *”Three decorative vases with minimalistic designs.”*

Fig. 10. Grad-CAM and LIME for hot pot (a) and vase (b) images, both explaining the named class of the same ResNet50. LIME tints each superpixel by its weight, dark red raising and blue lowering the probability of the class; Grad-CAM uses the jet scale, with red highest. *Images source: ImageNet dataset [4].*

7.3 User study

To assess the interpretability of visual explanation formats, we conducted a human evaluation involving 270 participants through a web-based survey, which was available in three languages: English, Czech, and Ukrainian. Each participant viewed 20 images in total—10 with bounding boxes and 10 with Grad-CAM heatmaps—and rated each on how well the display captures the named object, on a five-level scale from Very Poor to Excellent. At the end of the survey each participant also rated how easy the bounding-box explanations and the heatmaps were to comprehend overall. To minimize order bias, the images were shown in a randomized sequence.

The images presented to each participant were randomly selected from a pool of 100 classifier visualizations for each of the ten object classes. Each class was represented by exactly one pair of images per participant: one displaying a Grad-CAM heatmap and one showing a bounding box explanation. Figure 11 provides example questions used in the survey.



(a) *Image evaluations (heatmap)* (b) *Image evaluations (bounding box)*

Fig. 11. Illustrative examples of evaluation questions

Participants were also grouped by their self-assessed level of machine learning expertise. In addition to interpretability ratings, we collected the time it took participants to respond and also gathered open-ended feedback. This feedback was analyzed using GPT-4o to identify common themes and sentiment.

Table 10 summarizes the results. The survey asked two different questions, and they measure different things. At the end of the survey each participant rated how easy each format was to comprehend overall. Bounding boxes were rated significantly higher (mean 4.06 against 3.49; Wilcoxon $W = 2397.0$, $p = 8.0 \times 10^{-16}$; paired $t = 9.28$, $p < 0.001$). For each of the 20 images, participants rated how well the display captures the named object. Pooled over all ratings the two formats do not differ (means 3.139 and 3.131; Wilcoxon $p = 0.68$). By class, with Bonferroni-corrected Wilcoxon tests, bounding boxes were rated significantly higher for ambulance, acoustic guitar and hot pot, Grad-CAM for ant, bee and violin, and the other four classes show no significant difference. The Grad-CAM heatmaps were computed for the class predicted by ResNet50, which was the named class for 84.5% of the 1,000 images.

To enable a clear and consistent quantitative analysis, two types of participant ratings—overall comprehensibility and per-image object capture—were converted from text to numeric values using predefined mappings. For comprehensibility, the responses were encoded as follows: “Very easy to comprehend” = 5, “Easy to comprehend” = 4, “Comprehensible” = 3, “Difficult to comprehend” = 2, and “Very difficult to comprehend” = 1. For object capture, the mapping was: “Excellent” = 5, “Good” = 4, “Fair” = 3, “Poor” = 2, and “Very Poor” = 1.

Open-ended feedback was optional; 187 participants provided comments for bounding boxes and 187 for heatmaps. GPT-4o was asked to summarize each set of comments into short opinion topics, each with a sentiment and the number of participants who mentioned it. These are counts of topic mentions, not of participants, because a comment can mention several topics: 237 mentions for bounding boxes and 102 for heatmaps. Of the mentions, 73% were positive for bounding boxes (174 of 237) and 41% for heatmaps (42 of 102). Mentions from one participant are not independent

observations, so we report these proportions descriptively and do not test them. The themes identified using GPT-4o show that participants appreciated the clarity, accuracy, and simplicity of bounding boxes, while heatmaps were often criticized for being visually confusing, having overlapping colors, and lacking clear boundaries.

The preregistration of our study—including hypotheses, exclusion criteria, and analysis plan—has been omitted from the anonymized review manuscript and will be provided after review. Any analyses not included in the preregistration will be clearly marked as exploratory in the final report.

Table 10. Summary of evaluation results from the user study (N=270) comparing bounding boxes and heatmaps across interpretability and performance aspects.The p-value in the last column corresponds to the test type listed in the first column; better results for each pair are highlighted in bold.

Evaluation Aspect	**Bounding Box**	**Heatmap**	**Significance (p-value)**
Interpretability rating - one rating per participant at the conclusion of the survey (Wilcoxon test)	**Very easy: 89**	Very easy: 35	
	Easy: 112	Easy: 91	
	Comprehensible: 66	Comprehensible: 116	Yes ($p < 0.001$)
	Difficult: 3	Difficult: 27	
	Very difficult: 0	Very difficult: 1	
Overall comprehensibility rating (t-test)	**Mean = 4.06**	Mean = 3.49	Yes ($p < 0.001$)
How well the display captures the named object, by class (Wilcoxon test, Bonferroni-corrected)	**Ambulance: 4.06**	Ambulance: 3.36	Yes ($p < 0.001$)
	Fire Truck: 3.62	Fire Truck: 3.49	No
	Ant: 2.84	**Ant: 3.48**	Yes ($p < 0.001$)
	Bee: 2.76	**Bee: 3.68**	Yes ($p < 0.001$)
	Acoustic guitar: 3.33	Acoustic guitar: 2.97	Yes ($p < 0.001$)
	Violin: 2.96	**Violin: 3.37**	Yes ($p < 0.001$)
	Hotpot: 3.10	Hotpot: 2.70	Yes ($p < 0.001$)
	Vase: 3.23	Vase: 3.02	No
	Cucumber: 2.76	Cucumber: 2.71	No
	Zucchini: 2.73	Zucchini: 2.53	No
Open feedback: topic mentions by sentiment (descriptive)	**Positive: 174**	Positive: 42	
	Neutral: 9	Neutral: 5	Not tested
	Negative: 54	Negative: 55	
Open feedback themes	Clarity, precision, ease	Visibility, overlap, color confusion	Yes (qualitative)

8 Discussion

TEBOL's accuracy does not rest on the captioner alone. Against baselines that use the same images without the description stage, the pipeline reaches 0.980 against 0.952 for a CLIP linear probe, 0.949 for a CoCa linear probe, and 0.826 for matching the class name in the text. Zero-shot prompting of the captioner reaches 0.967, so the description stage is not what produces the accuracy; what it produces is a representation in which each word's contribution can be read off exactly. The primary contribution of this work lies in that interpretability rather than in maximizing classification accuracy. Even if the LLM recognizes the object class directly (e.g., generating the word 'ambulance'), the TEBOL pipeline ensures that this decision is transparently attributed to that specific semantic concept and can be semantically grounded in the image for inspection, unlike black-box vision models where the reasoning remains opaque.

What distinguishes the method is that its attribution is exact rather than approximate. Reconstructing a description's logit from its per-word contributions agrees with the model's own logit to 4.98×10^{-7} in the binary case and 5.01×10^{-6} across three classes, so the explanation is a decomposition of the model rather than a surrogate fitted to it. It also costs one dot product per word and returns the same ranking on every run, where a perturbation method needs thousands of samples and a random seed.

Removing the class name and its synonyms costs 0.123 to 0.239 of accuracy on four of the five tasks and nothing on hot pot / vase, so the class term does carry much of the signal. What remains without it, between 0.689 and 0.998, is still far above chance, so the classifier is not only reading the label out of the text. The captioner is the dominant source of variation in the explanation: five descriptions of one photograph share 0.640 of their words, against overlaps above 0.95 when the classifier is refitted or the explainer reseeded. The attribution contributes no variance of its own, which means the stability of a deployed system is governed by the captioner rather than by the attribution method.

From the perspective of applied perception, the broader significance of TEBOL lies in the structure of the explanation it produces. Instead of returning only a diffuse saliency artifact, the pipeline yields discrete semantic concepts and grounded regions that are easier to parse as coherent objects. Our findings therefore support the view that explanation quality depends not only on computational faithfulness, but also on whether the visual form of the explanation aligns with human perceptual organization.

The pipeline has two distinct explanatory roles. The textual attribution provided by SMER is faithful to the logistic regression classifier because it is analytically derived from the model parameters. In contrast, the bounding boxes are not faithful spatial explanations of that classifier. They are produced by an external multimodal LLM that grounds the SMER-selected concepts in the image. For this reason, our comparisons with Grad-CAM and LIME should be interpreted as comparisons of explanation formats and human interpretability, not as evidence that all methods reveal the same internal decision process.

9 Limitations

The proposed method faces challenges when dealing with abstract concepts, such as "love", for which a bounding box cannot be clearly defined. The same applies to words that name an attribute or a setting rather than an object: 4.6% of the words ranked among an image's three most important returned no coordinates at all, a rate that reaches 43.1% for *fire* and 23.5% for *guitar*. However, this limitation also affects most existing visualization methods, such as heatmaps, which similarly struggle to represent non-physical or abstract concepts.

While TEBOL leverages powerful multimodal LLMs for concept generation and localization, we emphasize that the pipeline's strongest interpretability claim rests on the faithfulness of the simple logistic regression classifier's decision-making process, which is inherently transparent and fully explainable via SMER. Our design therefore separates the opaque aspects of LLM generation from the interpretable aspects of classification. In particular, the bounding boxes should be understood as post-hoc semantic grounding supplied by an external model, not as faithful spatial explanations of the classifier itself.

The pipeline is agnostic to the choice of model. The two configurations reported here use a closed-weight model (GPT-4o with `text-embedding-ada-002`) and an open-weight one (Qwen3-VL-32B-Instruct with Qwen3-Embedding-4B), and the ordering of the methods is the same under both. All model settings and prompts are documented, and the generated descriptions, embeddings and predictions are provided.

Although our experiments focused on individual words, the method can potentially be extended to handle multi-word phrases. For example, this could involve using phrase chunking to detect expressions consisting of multiple words, followed by generating a single bounding box for the entire phrase.

In addition, the article discusses possible downstream uses of the representation, but does not implement or evaluate a full interface around it. Such scenarios should therefore be understood as design implications and future directions rather than as empirically validated interface claims.

We also observed that current LLMs may sometimes detect only part of a concept when it appears multiple times in an image. For instance, in images showing several vases, the LLM might draw a bounding box around only one or a few of them. The localization evaluation shows this: mean intersection over union falls from 0.845 on images with one annotated object to 0.734 with two and 0.588 with three or more.

A methodological limitation of this study is the relatively limited scale of the experiments. While our research utilized 14,351 images from ten object classes, this represents only a small fraction of the full ImageNet dataset, which contains over 14 million images across 20,000 classes. To fully assess the generalizability and robustness of the TEBOL method, additional benchmarking should be conducted on a broader variety of datasets and classification tasks.

A significant consideration when employing large language models trained on vast internet-scale datasets is the potential for data contamination or overlap with commonly used benchmarks such as ImageNet. While our experimental setup adhered to standard train/test splits for the ImageNet subset, we acknowledge the concern that the captioner might have been exposed to subsets of these images or similar concepts during its pre-training phase, potentially influencing the generated descriptions and thus the reported classification performance.

Section 6.2 measures one component of this directly: descriptions name their own class in a large share of cases, and removing the class name and its synonyms costs 0.123 to 0.239 of accuracy while leaving it far above chance. Contamination of the pre-training data remains untested, and datasets verified to lie outside it would be needed to settle it.

In the GPT-4o configuration, the average cost of processing a single image was approximately $0.0028, computed as the cumulative cost across all stages of the TEBOL pipeline. Token usage was aggregated over multiple API calls and models, including one multimodal GPT-4o call for image description, multiple embedding calls at the word level, and one multimodal GPT-4o call for bounding box generation. On average, the image caption contained seven words, resulting in seven embedding calls per image. The reported average of 971 tokens per image therefore represents the total number of tokens consumed across all pipeline components, including both textual inputs and internally encoded image representations in multimodal requests. Under this configuration, image description, embedding generation, and

bounding box prediction contributed approximately $0.0012, $0.000002, and $0.0016 to the total cost, respectively. The Qwen3-VL configuration runs locally and incurs no per-image API cost.

10 Conclusion

This study demonstrates that shifting the interpretability paradigm from pixel-level saliency to natural language concepts can enhance human understanding of image classifiers. By introducing the TEBOL pipeline, we showed that deterministic linear modeling applied to LLM-generated embeddings can yield word-level explanations that reconstruct the model's own logit to within 5×10^{-7}, while reaching 0.980 accuracy against 0.952 for the strongest baseline that does not pass through language. The ranked words survive removal of the class term, localize the annotated object with a mean intersection over union of 0.829, and are reproduced identically on every run, where a perturbation-based explainer requires thousands of samples and a random seed. SMER and LIME are equally faithful in both a closed-weight and an open-weight configuration, and the decomposition extends unchanged to three classes. Our quantitative evaluations and a user study with 270 participants show that participants rated concept-grounded bounding boxes as easier to comprehend overall than traditional heatmaps, while their ratings of how well each display captures the object are mixed by category, and clarify that these boxes represent post-hoc semantic grounding rather than the classifier's own spatial attention.

Ultimately, this work illustrates how multimodal alignment can support explainable computer vision by tracing predictions back to explicit, human-readable semantics. More importantly for applied perception, the results suggest that explanations become more usable when they are organized around discrete, object-centered structures that match how people visually parse scenes. As multimodal architectures continue to evolve, integrating perceptually aligned and language-grounded pipelines may therefore help bridge computational transparency and human understanding in high-stakes domains.

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References

[1] José P. Amorim, Pedro H. Abreu, João Santos, Marc Cortes, and Victor Vila. 2023. Evaluating the faithfulness of saliency maps in explaining deep learning models using realistic perturbations. *Information Processing & Management* 60, 2 (March 2023), 103225. doi:10.1016/j.ipm.2022.103225

[2] Dilip Arumugam, Mark K. Ho, Noah D. Goodman, and Benjamin Van Roy. 2023. Bayesian reinforcement learning with limited cognitive load. doi:10.48550/arXiv.2305.03263

[3] Christopher Burger, Lingwei Chen, and Thai Le. 2023. "Are your explanations reliable?" Investigating the stability of LIME in explaining text classifiers by marrying XAI and adversarial attack. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, Houda Bouamor, Juan Pino, and Kalika Bali (Eds.). Association for Computational Linguistics, Singapore, 12831–12844. doi:10.18653/v1/2023.emnlp-main.792

[4] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. 2009. ImageNet: A large-scale hierarchical image database. In *2009 IEEE Conference on Computer Vision and Pattern Recognition*. 248–255. doi:10.1109/CVPR.2009.5206848

[5] Jiarui Duan, Haoling Li, Haofei Zhang, Hao Jiang, Mengqi Xue, Li Sun, Mingli Song, and Jie Song. 2024. On the evaluation consistency of attribution-based explanations. doi:10.48550/arXiv.2407.19471

[6] Lucie Dvorackova, Marcin P. Joachimiak, Michal Cerny, Adriana Kubecova, Vilem Sklenak, and Tomas Kliegr. 2025. Explaining word embeddings with perfect fidelity: Case study in research impact prediction. doi:10.48550/arXiv.2409.15912

[7] Joakim Edin, Andreas Geert Motzfeldt, Casper L. Christensen, Tuukka Ruotsalo, Lars Maaløe, and Maria Maistro. 2025. Normalized AOPC: Fixing misleading faithfulness metrics for feature attribution explainability. doi:10.48550/arXiv.2408.08137

[8] Alex Fang, Gabriel Ilharco, Mitchell Wortsman, Yuhao Wan, Vaishaal Shankar, Achal Dave, and Ludwig Schmidt. 2022. Data Determines Distributional Robustness in Contrastive Language Image Pre-training (CLIP). arXiv:2205.01397 [cs.CV] https://arxiv.org/abs/2205.01397

[9] Timo Freiesleben, Gunnar König, Christoph Molnar, and Alvaro Tejero-Cantero. 2024. Scientific inference with interpretable machine learning: Analyzing models to learn about real-world phenomena. *Minds and Machines* 34, 3 (July 2024), 32. doi:10.1007/s11023-024-09691-z

[10] Luca Longo, Mario Brcic, Federico Cabitza, Jaesik Choi, Roberto Confalonieri, Javier Del Ser, Riccardo Guidotti, Yoichi Hayashi, Francisco Herrera, Andreas Holzinger, Richard Jiang, Hassan Khosravi, Freddy Lecue, Gianclaudio Malgieri, Andrés Páez, Wojciech Samek, Johannes Schneider, Timo Speith, and Simone Stumpf. 2024. Explainable artificial intelligence (XAI) 2.0: A manifesto of open challenges and interdisciplinary research directions. *Information Fusion* 106 (June 2024), 102301. doi:10.1016/j.inffus.2024.102301

[11] Sachit Menon and Carl Vondrick. 2022. Visual classification via description from large language models. doi:10.48550/arXiv.2210.07183

[12] George A. Miller. 1956. The magical number seven, plus or minus two: Some limits on our capacity for processing information. *Psychological Review* 63, 2 (1956), 81–97. doi:10.1037/h0043158

[13] Miquel Miró-Nicolau, Antoni Jaume-i Capó, and Gabriel Moyà-Alcover. 2024. Assessing fidelity in XAI post-hoc techniques: A comparative study with ground truth explanations datasets. *Artificial Intelligence* 335 (Oct. 2024), 104179. doi:10.1016/j.artint.2024.104179

[14] Miquel Miró-Nicolau, Antoni Jaume-i Capó, and Gabriel Moyà-Alcover. 2025. A comprehensive study on fidelity metrics for XAI. *Information Processing & Management* 62, 1 (Jan. 2025), 103900. doi:10.1016/j.ipm.2024.103900

[15] Shuhan Qi, Zhengying Cao, Jun Rao, Lei Wang, Jing Xiao, and Xuan Wang. 2023. What is the limitation of multimodal LLMs? A deeper look into multimodal LLMs through prompt probing. *Information Processing & Management* 60, 6 (2023). doi:10.1016/j.ipm.2023.103510

[16] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. 2021. Learning transferable visual models from natural language supervision. doi:10.48550/arXiv.2103.00020

[17] Zahra Rahmatinejad, Toktam Dehghani, Benyamin Hoseini, Fatemeh Rahmatinejad, Aynaz Lotfata, Hamidreza Reihani, and Saeid Eslami. 2024. A comparative study of explainable ensemble learning and logistic regression for predicting in-hospital mortality in the emergency department. *Scientific Reports* 14, 1 (Feb. 2024), 3406. doi:10.1038/s41598-024-54038-4

[18] Marco Tulio Ribeiro, Sameer Singh, and Carlos Guestrin. 2016. "Why should I trust you?": Explaining the predictions of any classifier. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD '16)*. Association for Computing Machinery, New York, NY, USA, 1135–1144. doi:10.1145/2939672.2939778

[19] Ramprasaath R. Selvaraju, Michael Cogswell, Abhishek Das, Ramakrishna Vedantam, Devi Parikh, and Dhruv Batra. 2017. Grad-CAM: Visual explanations from deep networks via gradient-based localization. In *2017 IEEE International Conference on Computer Vision (ICCV)*. 618–626. doi:10.1109/ICCV.2017.74

[20] K. Sevvanthi, Sachit Ganapathy, Prasanth Penumadu, and K. T. Harichandrakumar. 2023. Comparing the predictive performance of a decision tree with logistic regression for oral cavity cancer mortality: A retrospective study. *Cancer Research, Statistics, and Treatment* 6, 1 (March 2023), 103. doi:10.4103/crst.crst_234_22

[21] Shafiullah Soomro, Asim Niaz, and Kwang Nam Choi. 2024. Grad++ScoreCAM: Enhancing visual explanations of deep convolutional networks using incremented gradient and score-weighted methods. *IEEE Access* 12 (2024), 61104–61112. doi:10.1109/ACCESS.2024.3392853

[22] Luca Surace, Michał Wernikowski, Cemil Tursun, Karol Myszkowski, Rafał K. Mantiuk, and Piotr Didyk. 2023. Learning GAN-based foveated reconstruction to recover perceptually important image features. *ACM Transactions on Applied Perception* 20, 2, Article 7 (2023), 23 pages. doi:10.1145/3583072

[23] Ping Wang and Heng Ding. 2024. The rationality of explanation or human capacity? Understanding the impact of explainable artificial intelligence on human-AI trust and decision performance. *Information Processing & Management* 61, 4 (July 2024), 103732. doi:10.1016/j.ipm.2024.103732

[24] Wanni Xu, You-Lei Fu, and Dongmei Zhu. 2023. ResNet and its application to medical image processing: Research progress and challenges. *Computer Methods and Programs in Biomedicine* 240 (Oct. 2023), 107660. doi:10.1016/j.cmpb.2023.107660

[25] Jiahui Yu, Zirui Wang, Vijay Vasudevan, Legg Yeung, Mojtaba Seyedhosseini, and Yonghui Wu. 2022. CoCa: Contrastive Captioners are Image-Text Foundation Models. arXiv:2205.01917 [cs.CV] https://arxiv.org/abs/2205.01917

[26] Yunxiang Zhang, Nan Wu, Connor Z. Lin, Gordon Wetzstein, and Qi Sun. 2024. GazeFusion: Saliency-guided image generation. *ACM Transactions on Applied Perception* 21, 4, Article 14 (2024), 19 pages. doi:10.1145/3694969

[27] Zhaoheng Zheng, Jingmin Wei, Xuefeng Hu, Haidong Zhu, and Ram Nevatia. 2024. Large language models are good prompt learners for low-shot image classification. In *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 28453–28462. doi:10.1109/CVPR52733.2024.02688

A Attribution faithfulness in the GPT-4o configuration

The GPT-4o configuration (GPT-4o descriptions at seven words and the `text-embedding-ada-002` word vectors they were embedded with, both stored from the original run) was passed through the analysis of Section 6.4: the classifier is refitted on the 25 folds, and SMER and LIME are compared by perturbing that classifier. SMER and LIME again give the same curve. Their integrated scores differ by less than 0.001 on each of the four tasks. Cucumber / zucchini is not rerun, because its stored word vectors are 300-dimensional rather than ada-002.

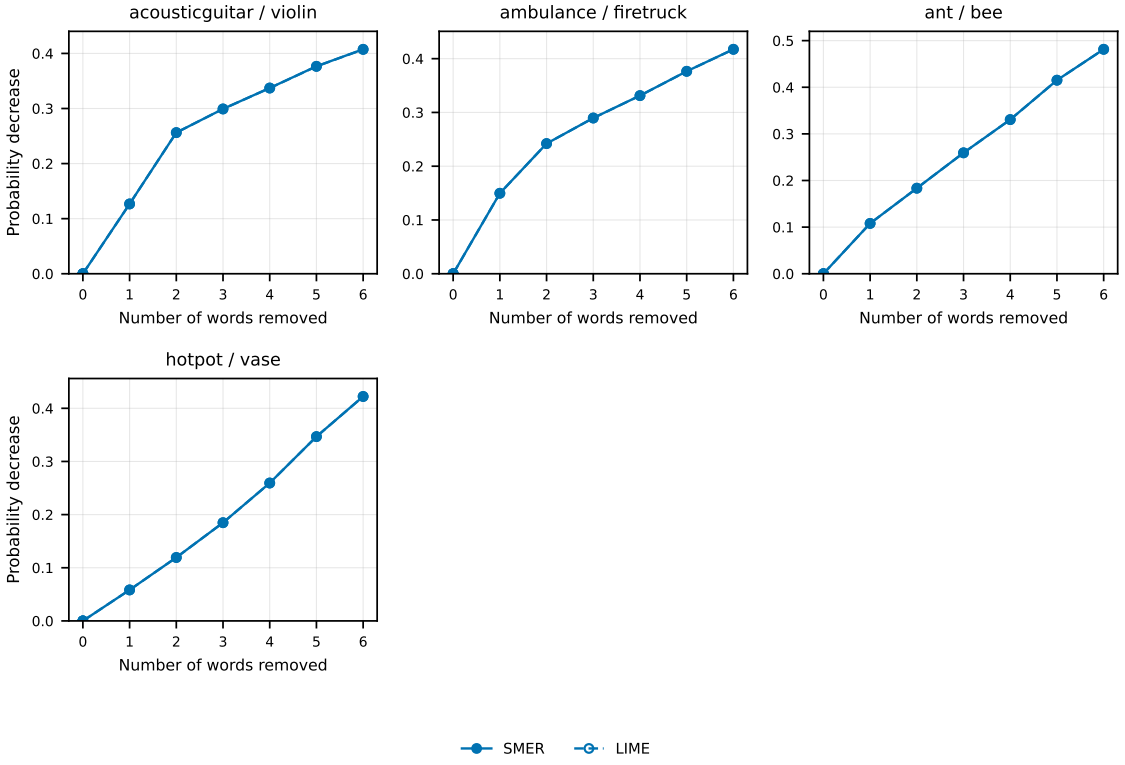


Fig. 12. Perturbation curves for the GPT-4o configuration, seven words. LIME coincides with SMER on every task, so its curves lie behind SMER's.

B Attribution stability: per-task values and caption length

Table 8 reports means over the five tasks at five words. Table 11 gives the underlying per-task values, and Table 12 repeats the measurement at seven and thirty words. Definitions and the direction of each measure are as in Section 6.5.

Table 11. Attribution stability per task, descriptions as generated at five words, Qwen3-VL configuration.

Source	Measure	guitar / violin	ambulance / firetruck	ant / bee	cucumber / zucchini	hotpot / vase
Refit, SMER	Jaccard@3	0.969	0.971	0.973	0.901	0.989
	Top-1	0.003	0.000	0.005	0.002	0.005
	τ	0.862	0.925	0.938	0.737	0.943
Refit, LIME	Jaccard@3	0.960	0.969	0.970	0.895	0.981
	Top-1	0.000	0.003	0.002	0.004	0.007
	τ	0.878	0.901	0.931	0.735	0.933
Reseed, LIME	Jaccard@3	0.952	0.982	0.977	0.987	0.989
	Top-1	0.000	0.000	0.003	0.002	0.002
	τ	0.848	0.957	0.957	0.959	0.950

Cucumber / zucchini is the least stable task under refitting for both methods ($\tau = 0.737$ for SMER and 0.735 for LIME), and it is also the task with the lowest classification accuracy. Hot pot / vase is the most stable.

Table 12. Attribution stability by requested description length, Qwen3-VL configuration. Mean ± standard deviation over the five tasks. The captioner row is the overlap of the three highest-ranked words across five regenerated descriptions; its word sets differ between runs, so the order-based measures do not apply.

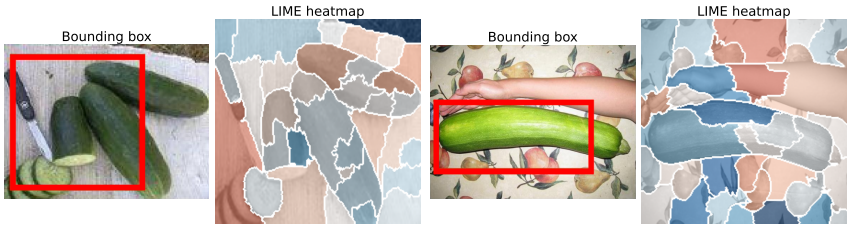
Length	Source	Jaccard@3	Top-1 changed	τ
5 words	Refit, SMER	0.961 ± 0.034	0.003 ± 0.002	0.881 ± 0.087
	Refit, LIME	0.955 ± 0.034	0.003 ± 0.002	0.876 ± 0.082
	Reseed, LIME	0.977 ± 0.015	0.001 ± 0.001	0.934 ± 0.048
	Captioner	0.663 ± 0.056	—	—
7 words	Refit, SMER	0.957 ± 0.032	0.003 ± 0.004	0.852 ± 0.061
	Refit, LIME	0.955 ± 0.027	0.006 ± 0.007	0.841 ± 0.064
	Reseed, LIME	0.974 ± 0.011	0.002 ± 0.003	0.899 ± 0.056
	Captioner	0.660 ± 0.060	—	—
30 words	Refit, SMER	0.939 ± 0.057	0.037 ± 0.040	0.543 ± 0.192
	Refit, LIME	0.943 ± 0.055	0.037 ± 0.039	0.517 ± 0.179
	Reseed, LIME	0.954 ± 0.022	0.039 ± 0.023	0.475 ± 0.051
	Captioner	0.545 ± 0.142	—	—

Stability falls with length, and it falls fastest in the tail of the ordering. From five to thirty words the top-three overlap under refitting drops by 0.022 for SMER, while τ drops by 0.338; the leading word changes in 3.7% of runs rather than 0.3%. The two methods diverge at thirty words: LIME holds the higher top-three overlap, 0.943 against 0.939, and SMER the higher full ordering, 0.543 against 0.517.

The captioner degrades over the same range. Five descriptions of one photograph share 0.640 of their words at five words and 0.462 at thirty, and the share of images receiving five identical descriptions falls from 19–28% to zero. Predictions follow: images receiving the same class from all five descriptions fall from 94.6–99.6% at five words to 75.8–99.2% at thirty. These results are the basis for reporting at five words in Section 6.5.

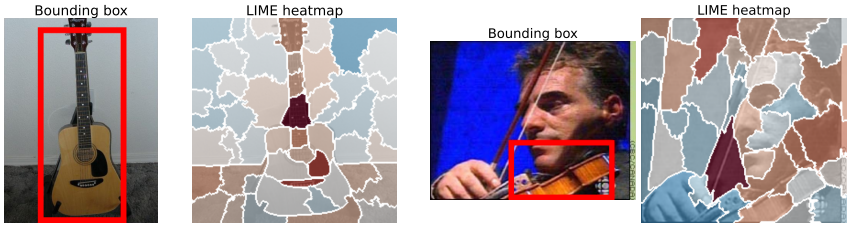
C Additional qualitative comparisons

Figures 13–16 give the TEBOL and LIME comparison for the four class pairs omitted from Section 7, and Figure 17 the LIME and Grad-CAM comparison for the vase image omitted from Section 7.2. All were produced with the GPT-4o configuration at seven-word descriptions.



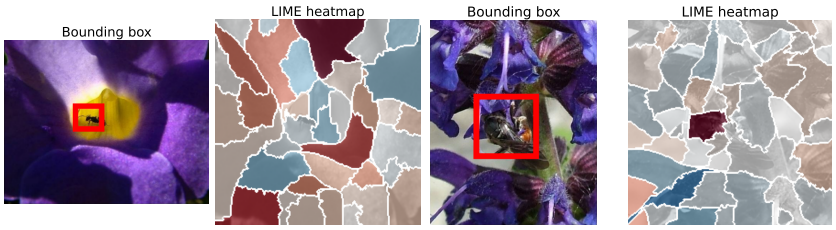
(a) *"Cucumber, slice cucumber and knife on a cutting board."* (b) *"Hand next to a large green zucchini."*

Fig. 13. Examples of a bounding-box explanation generated by our TEBOL method (left) and a LIME explanation (right) for Cucumber vs. Zucchini classification; colours as in Figure 9. *Images source: ImageNet dataset [4].*



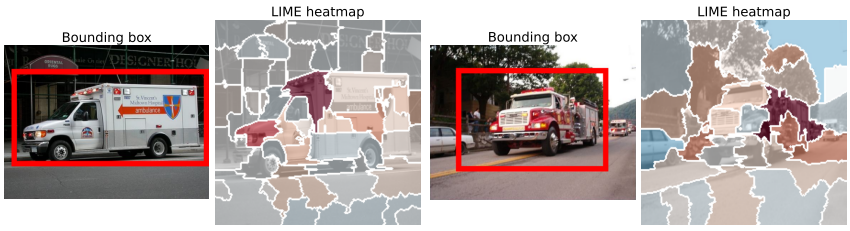
(a) *"Wooden acoustic guitar leaning against white wall."* (b) *"Man playing violin with focus on strings."*

Fig. 14. Examples of a bounding-box explanation generated by our TEBOL method (left) and a LIME explanation (right) for Guitar vs. Violin classification; colours as in Figure 9. *Images source: ImageNet dataset [4].*



(a) *"Tiny ant exploring inside bright purple flower."* (b) *"Bee perched on thorny stem in sunlight."*

Fig. 15. Examples of a bounding-box explanation generated by our TEBOL method (left) and a LIME explanation (right) for Ant vs. Bee classification; colours as in Figure 9. *Images source: ImageNet dataset [4].*



(a) *"White ambulance parked on a city street."* (b) *"Red fire truck driving down city street."*

Fig. 16. Examples of a bounding-box explanation generated by our TEBOL method (left) and a LIME explanation (right) for Ambulance vs. Fire Truck classification; colours as in Figure 9. *Images source: ImageNet dataset [4].*

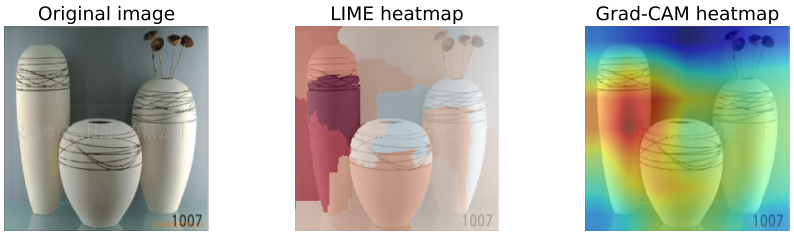


Fig. 17. LIME and Grad-CAM for a vase image, both explaining the class vase of the same ResNet50 classifier. *"Three decorative vases with minimalistic designs." Image source: ImageNet dataset [4].*